

HU 301: Engineering Economics (B.Tech IIIrd Yr.)

Details of the Course:

Course Title	Course Structure			Credits	Pre-Requisite	Relative Weightage		
	L	T	P			CWS	MTE	ETE
HU 301 Engineering Economics	3	0	0	03	Nil	25	25	50

Course Objective: To enable the students to understand the economic theories which may be applied to maximise return and economic environment in which they have to operate.

Course Outcome (CO):

1. Identify the core concepts of economics and recognise its potential in addressing modern day socio-economic challenges
2. Compare and analyse alternatives based on present, annual, rate of return, benefit over cost analyses, time value of money in evaluation of investments and projects in real life and the impact of economic factors on feasibility of real life projects.
3. Analyse and predict the economic impact of engineering solutions to make financially prudent decisions in everyday life.
4. Evaluate the role of economics To give knowledge to the students about various costs for determining the manufacturing of a product.
5. Collaborate on real projects and apply economic principles to alleviate socio-economic problems.

S. No	Content	Contact Hour
Unit 1	Introduction: Nature and significance of economics, Goods and Utility, Basic Concept of Demand and Supply, Elasticity of Demand- Price elasticity of Demand, Cross elasticity of Demand, Production - Production Function, Production Process and Factors of Production, Market - Introduction to Monopoly, Perfect Competition, Oligopoly and Monopolistic Competition, Cost Concepts, E-commerce.	10
Unit 2	Money- its evaluation and function, Bank- Commercial Bank and Central Bank and brief idea about function of banking system. Tax and Subsidy, Type of Tax- Direct and Indirect, Monetary and fiscal policy, Inflation and Business cycle, IPR & WTO, International trade, terms of Trade, Gain from International Trade, Free Trade vs. Protection, Dumping, and Balance of Payment	11

Unit 3	Role of Science, Engineering and Technology In economic Development: Some of the burning problems of rural and slum areas in India and how engineering and technology may be used to alleviate them, example of Green Revolution and White revolution. Reasons for their success and can we replicate them. Sustainable Development	12
Unit 4	Elementary Economic Analysis; Interest formulas and their Applications; Calculations of economic equivalence, Bases for Comparison of Alternatives: Present Worth Method, Future worth method, Annual equivalent, Internal Rate of return; Evaluating Production Operations, Business Risk Management.	12
	Total	45

Suggested Books:

S.No.	Name of Books/Authors/Publisher
1.	G.J. Thuesen, & W.J. Fabrycky, Engineering Economy, Prentice-Hall of India Private Limited, New Delhi.
2.	William G. Sullivan, James A. Bontadelli & Elin M. Wicks, Engineering Economy, Pearson Education Asia, First Indian reprint.
3.	Donald G. Newnan, Jerome P. Lavelle & ted G. Eschenbach, Engineering Economic Analysis, Engineering press, Austin, Texas.
4.	Seema Singh, Economics for Engineering Students, IK International Publishing House Pvt. Ltd.